

MeshAI

Run a model your hardware cannot hold, on the devices you already own.

30B across a laptop and two phones

ready in 70 s

6.6 tokens per second

internet off

What if you could actually run it?

no API bill

on your phones

30B coder

reviews your code

on a budget laptop

offline

Memory decides which model you can run.

WHAT A MODEL NEEDS

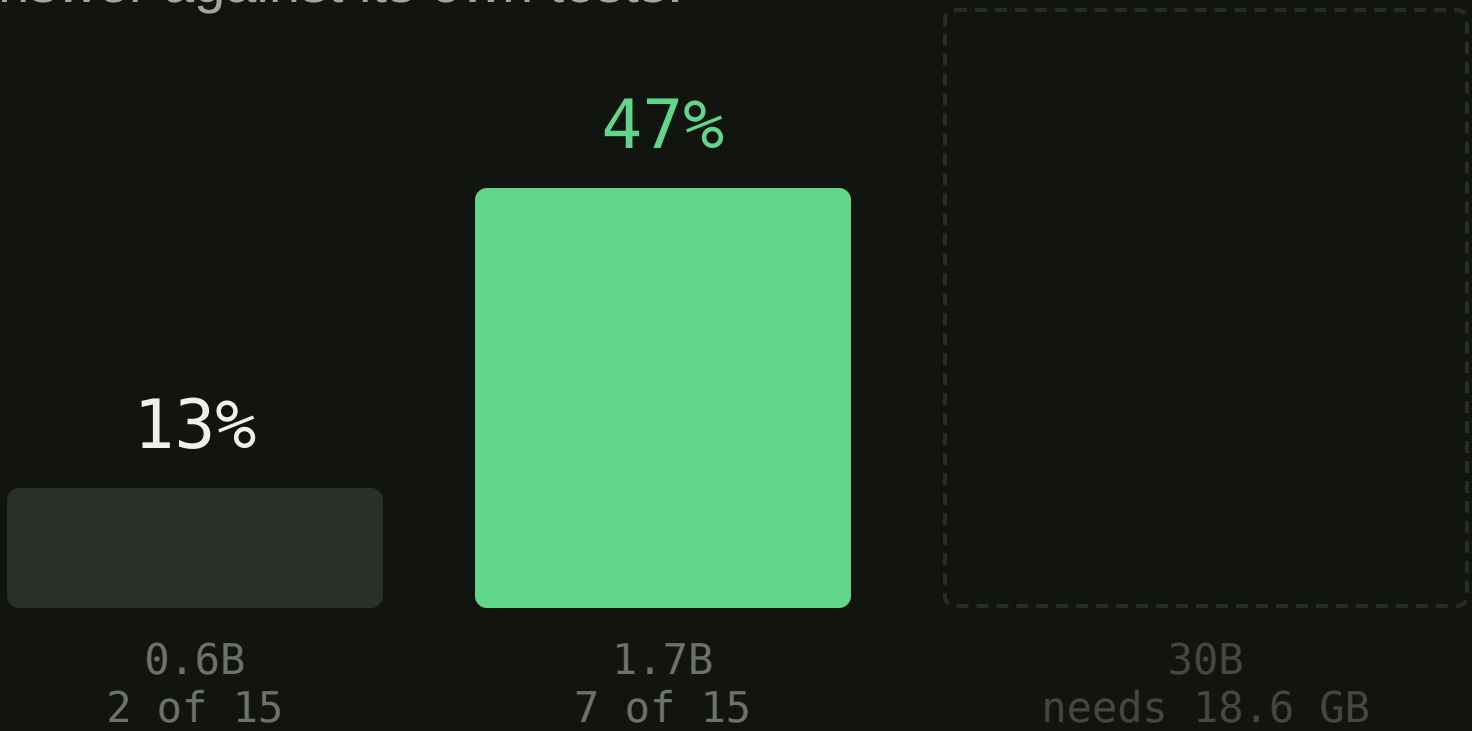
Qwen3 0.6B	0.64 GB
Qwen3 1.7B	1.83 GB
Qwen3 8B	4.7 GB
gpt-oss 20B	12.1 GB
Qwen3-Coder 30B	18.6 GB

WHAT A DEVICE HAS FREE

- A budget laptop: about 4 GB
- One 16 GB phone: about 8.7 GB
- The three together: about 20 GB
and the 30B fits

SMALL MODELS ARE NOT ALMOST AS GOOD

The same 15 coding problems, scored by running every answer against its own tests.



Neither device can hold the one that is worth running. A machine that can costs more than the laptop.

Pool the memory of the devices you already own.

- One device plans, the others hold layers
it reads free memory, heat, battery and link, then decides
- One OpenAI-compatible endpoint
the tools you already use do not change
- About 4 KB crosses the cable per word
the 18.6 GB of weights never move
- Any device, mixed: phones and laptops in one cluster, any GGUF
nothing hardcoded; we read the header and size it ourselves
- It refuses a device, in words, when it would make things worse
- Nothing leaves your devices. The demo runs with the internet off

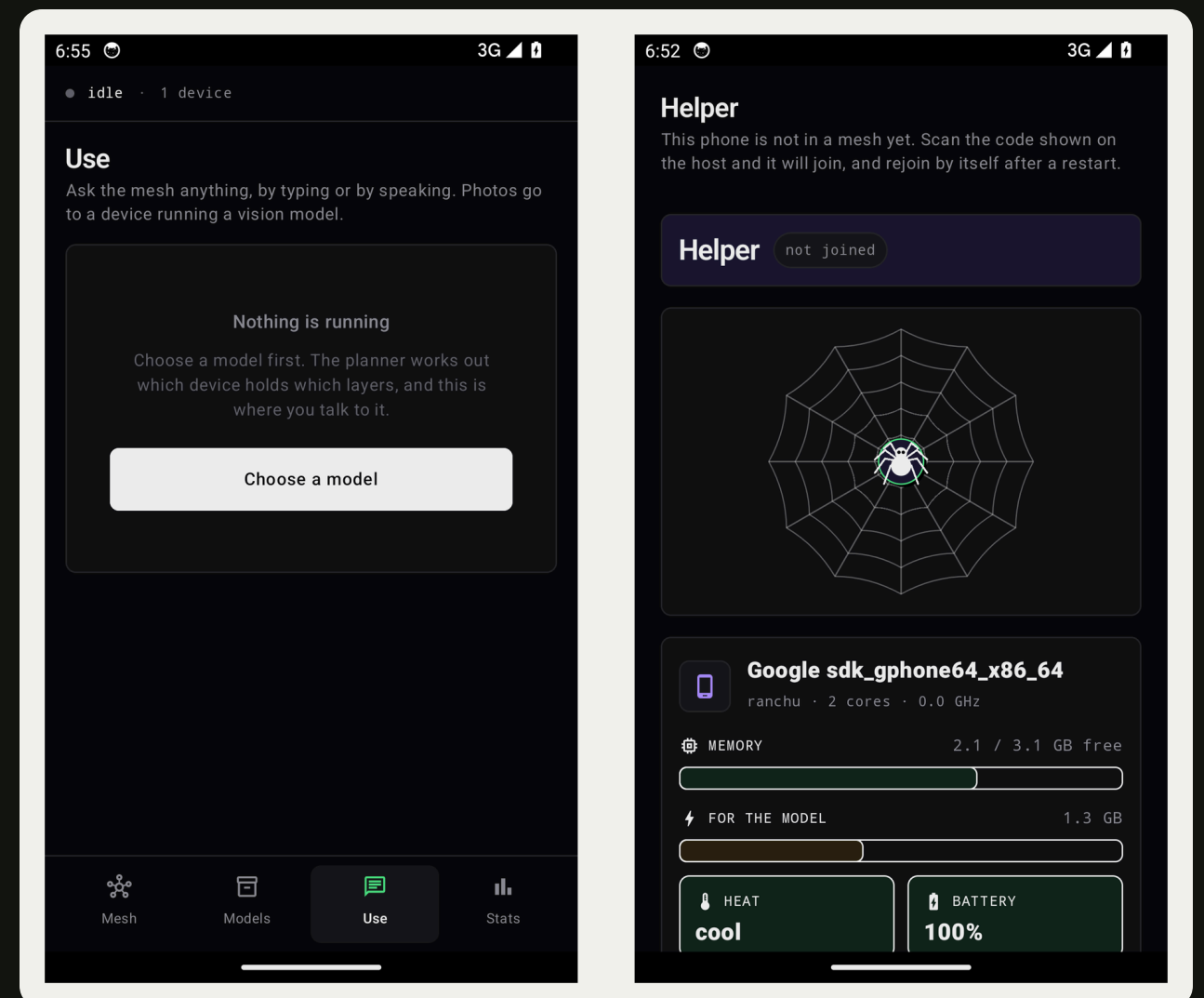
one endpoint

plans the split

refuses bad devices

mesh review · tests · diff

git pre-commit hook



You are about to commit. You tap one button.

YOUR LAPTOP

You edit a file
git diff, not committed



YOUR PHONE

One tap: review my
changes
it asks the laptop over
the cable



THE MESH

30B across three
devices
two phones hold 9.6 GB
of it



THE REVIEW

On the phone screen
before you commit

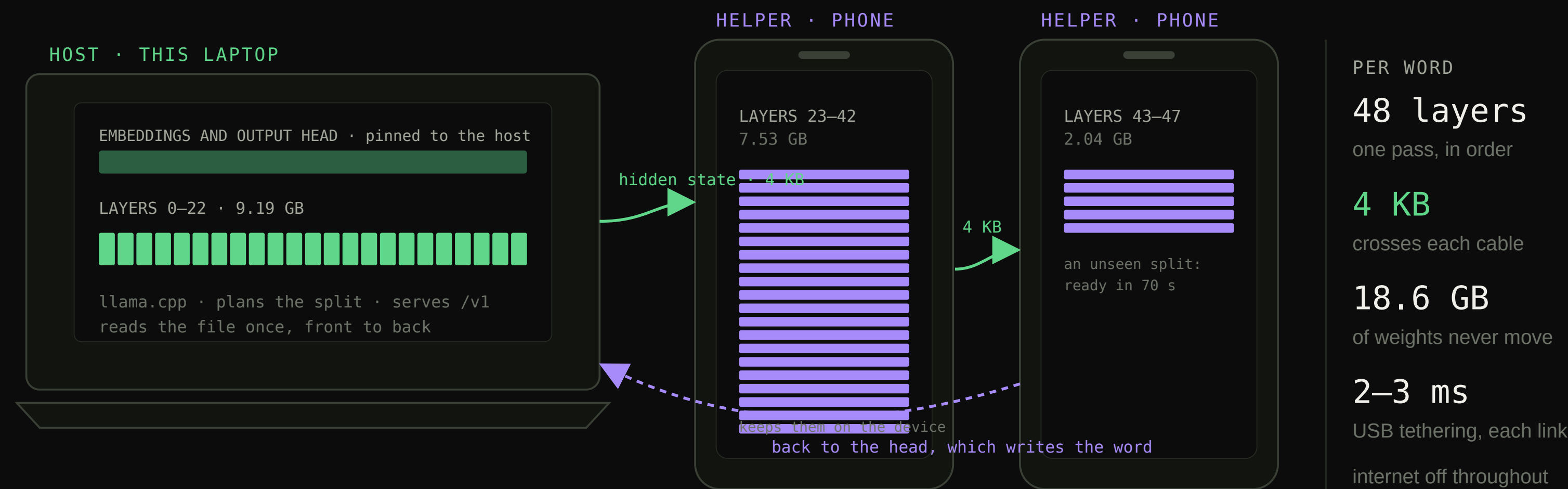
WHY ANYONE KEEPS IT

- The same review installs as a git pre-commit hook, in one command
- Every commit read before it lands, on your own hardware
- No API bill on the ten thousandth run

AND IT NEVER LEAVES

- Internet off · no account · no upload
- The code stays on the desk it was written on
- Also: mesh review a file, mesh tests to write its tests

One word, one pass, three devices.



THE PLANNER DECIDES

Live memory, heat, battery and round trip time. One unbroken run of layers each, largest device first.

THE HELPER NEVER SEES YOUR PROMPT

It receives a hidden state, computes its layers and hands the result back. No text ever reaches it.

THE HEAD STAYS HOME

Embeddings and output head are pinned to the host, which is why only kilobytes cross per word.

What made a split usable at all.

LOADING

517 s → **70** s

With mmap the engine read the file in scattered pieces while pushing layers out. Read it once, front to back.

A SPLIT NEVER USED BEFORE

437 s → **70** s

Every device keeps its own layers, checked against the engine's own file.

AND THE REST

- Embeddings and output head pinned to the host
so one small activation crosses per word, not weights
- One conversation slot
a long chat stays in the cache instead of being re-read
- Median of ten round trips, never the average
one Wi-Fi spike cannot evict a healthy device
- A sustained run, not a single shot
11.4 → 5.3 tok/s as the phone goes 34.6 → 47.5 °C, raw CSV committed
- Text and vision on two phones at once

IT REFUSES

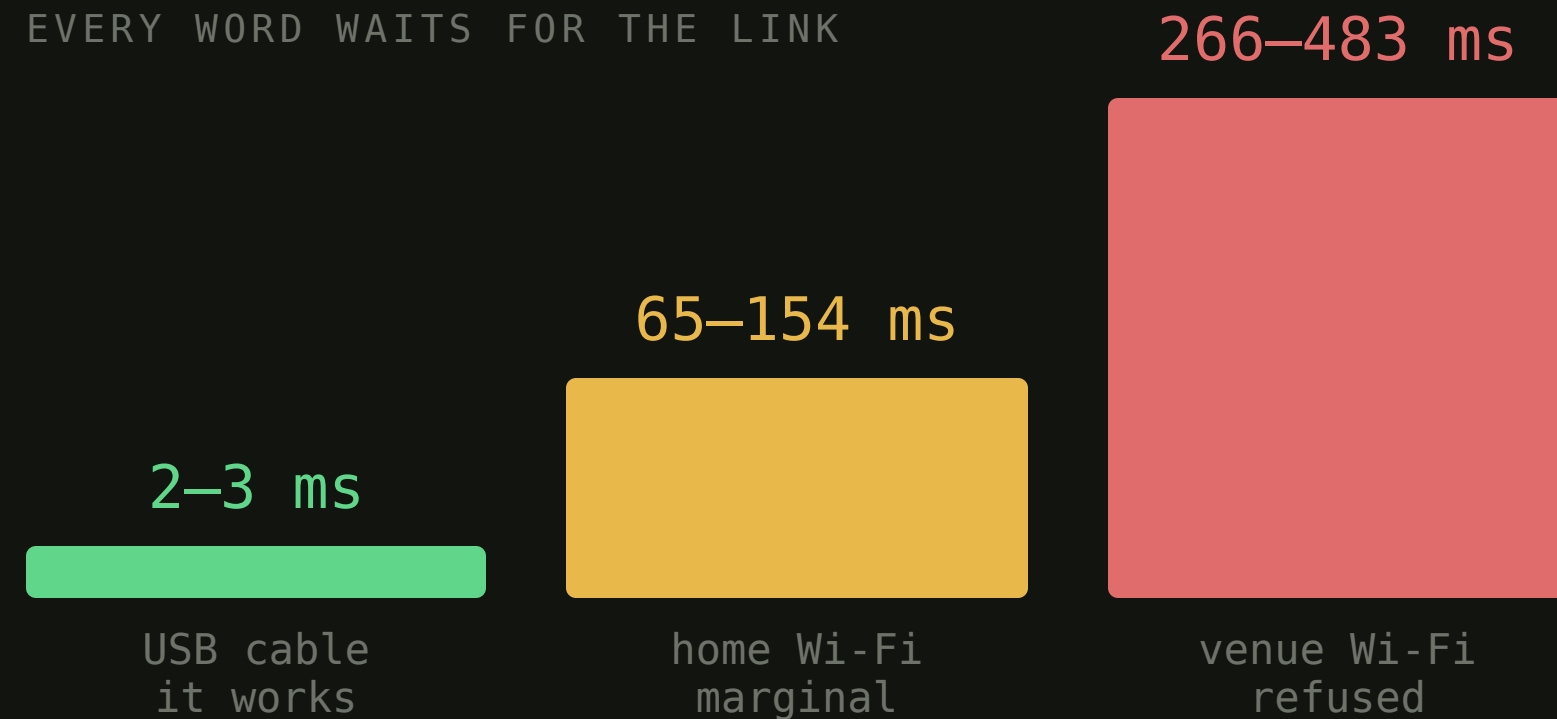
The planner says no, in a sentence you can read.

- Too slow to reach, above 60 ms
- Too hot, or nearly flat
- The model already fits one device
splitting would only make it slower

Seven hermetic tests cover the planner. It will tell you your phone is not worth using, and that is the feature.

The link decides everything.

EVERY WORD WAITS FOR THE LINK



Measured round trip. The planner refuses anything above 60 ms, which is why the demo runs on a cable.

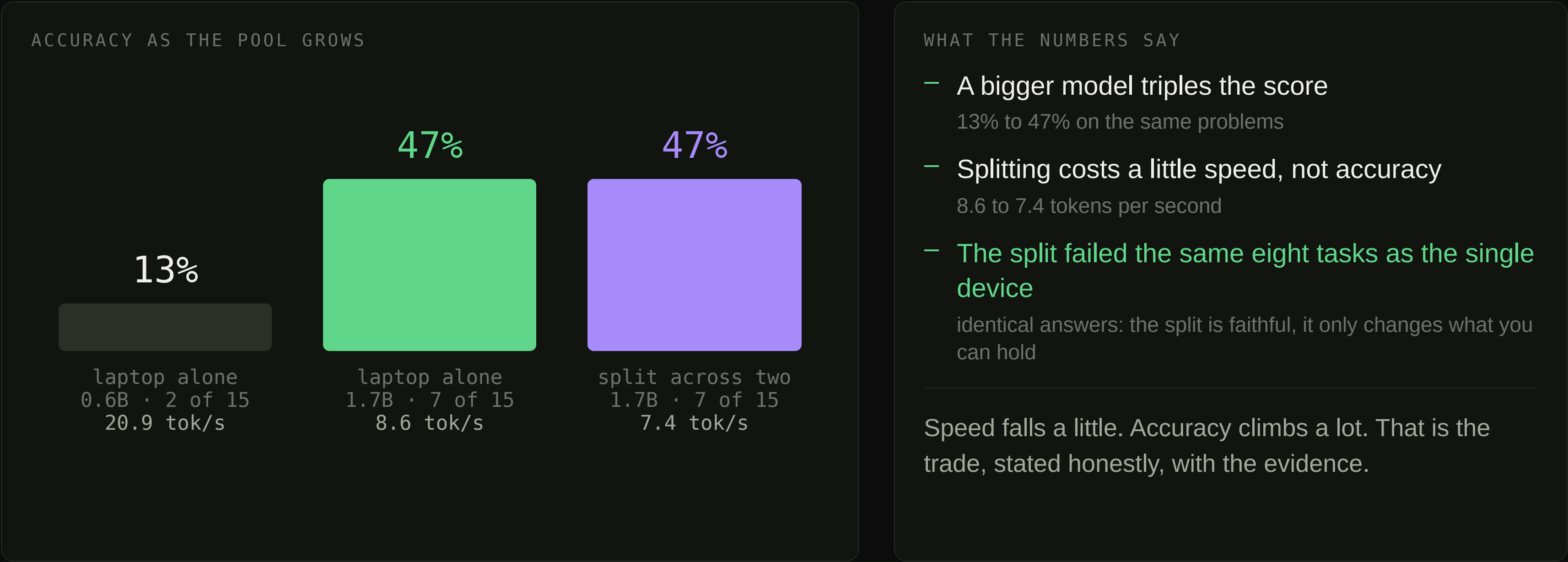
WHAT THE RESEARCH SAYS

- Petals measured 0.83 tokens per second across two continents
the only peer-reviewed system to do this at scale
- `prima.cpp`, the most mature home cluster, excludes phones on purpose
- A search of arXiv for multi-smartphone inference returns nothing

A split is worth it on a link you own,
and worthless on one you do not.
That is exactly where we drew the line.

Measured, not claimed.

The same fifteen coding problems, asked of every configuration, scored by executing each answer against its own tests.
No partial credit, no human judgement.



What already exists, and what it leaves undone.

SYSTEM	WHAT IT DOES	WHAT IT LEAVES TO YOU
Darkbloom Eigen Labs, a16z backed	Rents idle Apple Silicon Macs for inference and pays their owners	One whole model per Mac. It does not pool memory, and never touches a phone
exo github, 47k stars	Splits a model across your own Macs and Linux boxes, by how much memory each has	The phone is not a first class device. It does not refuse a bad one, or explain
Petals peer reviewed	A public swarm of volunteer servers holding one very large model	0.83 tokens per second across two continents. Unmaintained since 2024
llama.cpp RPC the primitive we use	Runs layers on another machine over a socket. MeshAI drives this underneath	You type the addresses and the tensor split by hand. Nothing decides
Ollama · LM Studio Jan, and the rest	Excellent single device runtimes. One machine, one model	If the model does not fit that machine, you do not run it at all
MeshAI what we built	An Android app where a phone hosts or helps. A planner reads live memory, heat, battery and link, gives each device layers it can hold, and says no in words	Nothing. No paper and no product pools phone memory to run one model

ANY DEVICE, MIXED

Android phones and laptops in one cluster,

THE PHONE IS A PEER, NOT AN ACCESSORY

It can be the host that plans and answers, or

ALL OF IT ON THE DEVICE

Camera into a vision model, voice

Their own devices become the private cloud.

We do not buy hardware and we do not rent anyone else's. We sell the software that turns the devices an organisation already owns into the cluster that runs its own AI.

WHO PAYS

- Teams whose code cannot leave the building
- Developers whose laptop is the limit
- Labs and campuses with devices idle all night
- Anywhere the network is poor or absent

They are not choosing between us and a cloud. They are choosing between us and a GPU server at eight to fifteen lakh, or nothing at all.

WHAT THEY GET

- A 30B model on the devices they own
- One endpoint their tools already speak
- Every commit reviewed locally
- Nothing leaves the building, ever
- No bill on the ten thousandth run

Add a device and the ceiling rises. What you can hold is what you can run.

HOW WE EARN

- **Per site**
a lab, a floor, a campus: the private cloud
- **Per seat**
the developer tool, with the git hook as the way in
- **Licensed to the device maker**
two phones worth more together than apart, which is a reason to own the second one

That last buyer is sitting in this room.

It is not a plan. It ran this weekend.

WHAT RAN

Qwen3-Coder-30B

- 18.6 GB across a laptop and two phones
- Laptop 0–22 · phone 23–42 · phone 43–47
the phones hold 9.6 GB of it
- Ready in 70 seconds · 6.6 tokens per second
- Three coding questions answered correctly
- Internet off throughout

Conditions, because a number without them is worthless:
USB tethering 2–3 ms · context 4096,

BUILT INSIDE THE WINDOW

- llama.cpp cross compiled for arm64
- Android app: host role and helper role
- Rust agent and its web panel
- Our own GGUF reader and layer planner
- Layer store, so an unseen split starts in 70 s
- Benchmark harness that runs the answers
- CLI and a git pre-commit hook

Every commit timestamped. Nothing pre-built. Open source used with attribution, all of

WHAT COULD GO WRONG, AND WHAT WE DID

The link is slow	refused above 60 ms, in words
Phone runs out of memory	proportional reserve, owner sets a cap
A device drops	it reconnects; a real departure ends cleanly
Engine lacks a flag	we ask the binary what it supports
A corrupt model file	every header read is bounds checked
An unfamiliar machine	one press checks engine, models, ports, adb

It refuses before it disappoints.

What we will not claim.

Volunteering these is what makes the rest believable.

NOT FASTER

It buys memory, not speed.

- Every word crosses the link once
- A split is slower than one device that already fits
- Our own planner says so, and refuses your phone
- We split only when the model fits nowhere else

NOT CHEAPER PER TOKEN

A commodity API is cheaper. That is not the point.

- We are for work that is not allowed to leave
- And for hardware that cannot hold the model alone
- If your data can leave and you want the best answer, use the cloud

PRIVATE, BUT PRECISELY

Across devices one person owns. No further.

- Nothing leaves the devices you own
- The demo runs with the internet off
- We do not claim activations sent to a stranger are private
published work turns them back into text

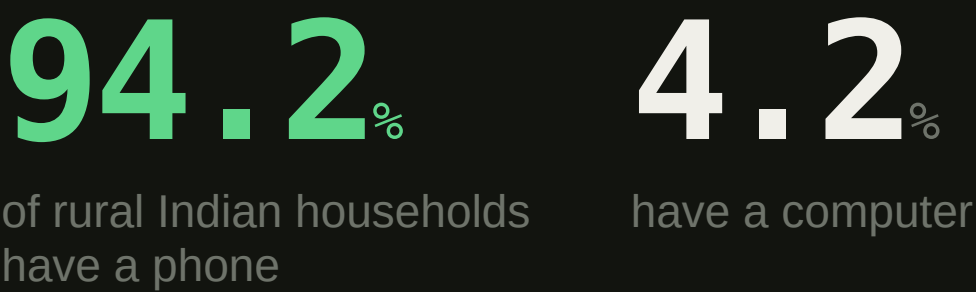
The ceiling moves.

WHO IT IS FOR

- developers on a weak laptop
- two person teams
- workplaces that forbid cloud AI
- students with a good phone
- field work with no signal
- any capable Android

- Memory decides which model you can run
- Pool the memory and the ceiling moves
- The same devices now hold a 30B
- The answers change accordingly

WHY PHONE FIRST IS NOT A GIMMICK HERE



That is not our first user, and we will not pretend it is. It is why the phone has to be the device that *works* rather than the accessory: for most people it is the only computer in the house, and as open models improve at a given size, what it can hold matters more every year.

MoSPI, Comprehensive Annual Modular Survey 2022-23

Tech stack.

THE RUNTIME

- Any GGUF model
we read the header and size it ourselves
- llama.cpp and ggml, with its RPC transport
MIT, attributed
- Our own planner and GGUF reader
seven hermetic tests

THE DEVICES

- Kotlin and Jetpack Compose
host role and helper role in one app
- Android NDK, arm64 with dot product and i8mm
- Foreground service, type connectedDevice

THE LINK AND THE TOOLS

- Rust agent and web panel
mesh host, mesh join, mesh ask
- USB tethering, 2 to 3 ms
- OpenAI-compatible API, CLI and git hook
- No cloud, no account, no telemetry

Thank you

A model your hardware should not be able to run, on devices you already own, with the internet switched off.

Prajwal Gunnala

github.com/prajwal-gunnala/maynards

Yuva Raj Ambati

every number in this deck is in that repository

30B across three devices

ready in 70 s

6.6 tokens per second

4 KB per word

internet off